

Database Tables



Database: customer_churn

db_customer
customerid
name
country
State
gender
dob
interests
pincode



db_subscription
customerid
subscription_start_date
subscription_type
renewal_date
plan_type
contract_type
cancellation_date
cancellation_reason
monthly_charges
cltv
churn_score



db_support
customerid
complaint_date
Escalations
csat_score
col_1
comment

Calculated metrics

KPI	Formula
churn rate	churned customer / total customers
Churn by plan type	churn rate GROUP BY plan_type (Basic/Standard/Premium)
Churn by state	churn rate GROUP BY country, state
Retention rate	1-churn rate
ARPU	SUM(monthly_charges) / COUNT(active customerid)
Average customer tenure	AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))
Revenue at risk	SUM(monthly_charges) WHERE churn_score > 70
Escalation rate	SUM(escalations) / COUNT(complaints) × 100
Avg complaints per customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Correlation: escalations → churn	churn rate WHERE escalations ≥ 1 vs 0

Portfolio Project - Summary

- **Churn & Revenue Impact** *(general summary + with numbers from dataset)*

Engineered an end-to-end churn analytics pipeline for an OTT subscription dataset (20+ KPIs), identifying a 28.6% overall churn rate and surfacing that monthly-contract subscribers churned at 55.6%, which is 6.7 times the 8.3% annual-contract rate- directly attributing \$73.94/mo in MRR leakage and \$2,047 in CLTV erosion to six at-risk customers, enabling a targeted contract-migration retention strategy.

Churn Analysis - Roadmap

- **Connect** SQL database to Python – **pandas** and **sqlite3**
- Data **import** using SQL query in Python – **pandas** and **sqlite3**
- Data **Cleaning** – **numpy** and **pandas**
 - data types, rename cols, select specific cols, QCs, handle missing/null values
- **Feature Engineering** – **numpy** and **pandas**
 - Create new calculated cols, data transformation, use filters
- Data **Analysis** – **numpy** and **pandas**
 - EDA - aggregation, group by, pivot table
- Data **Visualization** – **matplotlib** and **seaborn**
- **Insights** on ppt – *key insights, next steps, impact to show in resume*

